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Physics-Informed Neural Networks (PINNs) for Improving a Thermal Model in Stereolithography Applications

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Abstract

Stereolithography (SLA), additive manufacturing (3D printing) technique, is widely used nowadays for rapid prototyping and manufacturing (RP&M). This technique is driven by photo-polymerisation, which is an exothermal process. This may lead to thermal stresses significantly affecting the final quality of printed parts/products. To guarantee high-quality parts printed with the SLA technique, understanding the thermal behaviour is therefore crucial for optimizing the process. In this paper, the recent physics-informed neural network (PINN) methodology was employed to improve a physics-based model for predicting the thermal behaviour of SLA processes. The accuracy of the improved thermal model is demonstrated in this paper by comparing the predicted 2D temperature field with the 2D temperature field measured by a high-speed infrared thermal camera on parts printed on a production machine.

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Keywords: Physics-informed neural network (PINN); Stereolithography (SLA); Thermal model; High-speed thermal imaging

1. Introduction

Stereolithography (SLA) translates computer-aided design (CAD) models into solid objects through a combination of laser, resin, photo-chemistry and software technologies [1, 2]. There are 3 basic processing techniques of SLA, namely (1) *Bottom-Up Printing*, where resin is cured through a window in the bottom of the vat by a light source from below; (2) *Top-Down Printing*, where resin is cured by a light source above the vat and the build platform is lowered down into the resin vat; and (3) *Continuous Liquid Interface Production (CLIP)*, where resin is cured through an oxygen permeable window in the bottom of the vat by a light source from below. Graphical illustrations of the 3 different SLA processes can be seen in [2].

In general, the SLA process consists of the following steps. *In the first step*, a CAD model of a part to be made is generated using a CAD software tool. *In the second step*, a dedicated software slices the CAD model into a series of thin horizontal layers. *In the third step*, the sliced information is transferred to an ultraviolet (UV) laser that scans a layer of photosensitive resin to cure it. For top-down SLA machines, the laser scans the top layer, while the laser scans the bottom layer for Bottom-Up/CLIP SLA machines. *In the fourth step for the Top-Down SLA machines*, the newly built layer attached to the build platform is lowered to a depth of just one layer, and a new layer of resin is recoated on top of the previous one. Meanwhile, *in the fourth step for the Bottom-Up/CLIP SLA machines*, the newly built layer attached to the build platform is raised, and a new layer of resin is recoated below the previous one. The third and fourth steps are repeated layer-by-layer until the part is complete.

Nowadays, parts manufactured with the SLA technique are mainly used as temporary, such as lost wax material, model, prototype design and so on, which greatly limits the promotion and usage of the technology. As stated in [3], there are two major barriers, comprising (*i*) photosensitive resin material and (*ii*) the SLA processing technique itself, that need to be addressed to push the applications of SLA beyond the current boundary. In this paper, the authors made attempts to address one of the

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two major barriers, namely the SLA processing technique. This is addressed by developing a methodology to improve a thermal model that can be used as a virtual thermal measurement during printing processes. Accurate thermal monitoring and control of SLA processes allow high printing speed and high precision.

When the UV light hits the photosensitive resin, the curing process takes place, generating heat that will accumulate and increase the temperature of the printed area unless it is allowed to dissipate. Depending on the scanning strategy and the geometry of the part to be printed, the increased temperatures at different locations may not be similar. As a result, the temperature distribution in the printed part is uneven, yielding thermal stresses that will significantly affect the final quality of printed parts/products.

To maintain high-quality products and increase the printing speed, the temperature distribution during the process, therefore, needs to be monitored and controlled. Since high-resolution spatial-temporal temperature measurement is typically not available in production SLA machines, an accurate thermal model is required to predict the temperature distribution during production. To this end, the execution of the model must be sufficiently fast for the control of production. Several models to simulate the behaviour of printed parts during SLA processes have been reported in the literature [1, 4, 5, 6]. These models typically focus on the full SLA processes, not just the thermal behavior, and are solved with numerical techniques such as finite element methods. The computation time of such models is typically too long for their predictions to be used during production.

This paper focuses on estimating the temperature distribution during the cooling phase with a hybrid modelling approach with a structure motivated by the physics while the choice of its parameters is motivated by a data-driven approach using Physics Informed Neural Networks (PINNs, see Section 4). The temperature prediction during the scanning step is out of scope. Note that this paper does not present a substantial new research on PINNs itself but rather combines techniques that allow us to employ PINNs to derive a more accurate, yet computationally low cost, a physics-based model that can be used in production.

The remainder of this paper is structured as follows. Section 2 presents the experimental test setup used in this study. Section 3 presents an initial thermal model relevant to SLA processes. Section 4 presents a methodology to improve the initial thermal model based on PINNs. Section 5 discusses the numerical simulation results. Section 6 concludes the main findings of this work and proposes some future work.

2. Experiment

Figure 1 illustrates a top-down SLA process, which is the focus of this study. In addition to the standard SLA hardware, we mount a long-wave infrared (IR) thermal camera above the printing surface to record the temperature across the surface area and over time.

A test campaign was designed to print some parts with basic geometry like cylinders of different diameters and blocks.

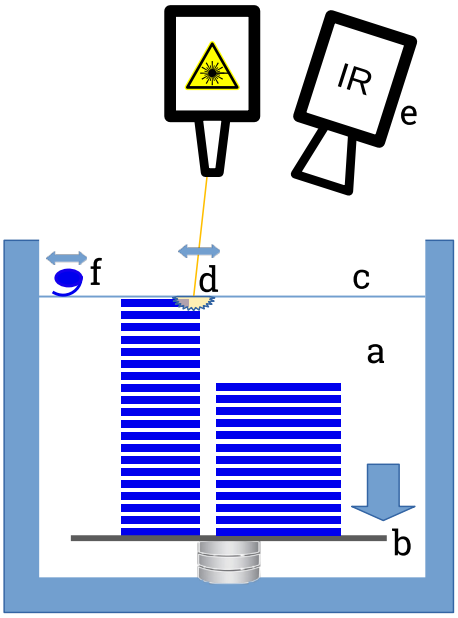


Fig. 1: SLA process monitored by thermal imaging. (a) photo-sensitive resin; (b) height-adjustable build platform; (c) resin surface; (d) laser-activated photo-polymerisation reaction; (e) IR thermal camera; (f) recoater arm.

During the production of the parts, the thermal images of the print area were recorded using the installed IR camera, time-synchronized with the building process. The thermal images were calibrated using a black-body reference surface over a wide temperature range. We also performed geometric calibration between the scanning field and the camera view. Figure 2 shows representative thermal images of the printed parts obtained from the test campaign where different printed parts are visible.

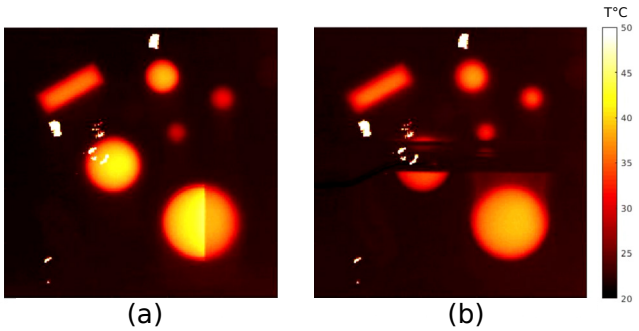


Fig. 2: Experimental data sample. An infrared camera placed above the system captures images of the temperature during an SLA process. In (a) the cylinder at the bottom, the laser beam is triggering photo-polymerisation, an exothermic reaction that leads to an elevation of the temperature. In (b), the recoater passes in front of the camera obstructing the lens. The recoater also provokes in-plane diffusion (see cylinder borders at the bottom). The white spots are due to defects of the IR camera lens.

Figure 3 shows the temperature at one point on the surface throughout printing a layer. Most of the heat is generated quickly after the laser passes by. To avoid excessive temperature built-up, the printer has to pause in between layers, allowing the part to cool down for longer than the time needed for the laser

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to scan a layer. Waiting unnecessarily long reduces the productivity of the machine, and not waiting long enough reduces the product quality. It is therefore important to model the cool down process accurately. It consists of three phases. (1) During the initial cool down, the solid material is exposed to air above. (2) When the recoater wipes over the surface, it deposits a new thin layer of liquid resin with a different temperature over the solid layer. (3) Between the recoating and the next laser pass at the location, the thin liquid layer is warmed by the solid object just underneath.

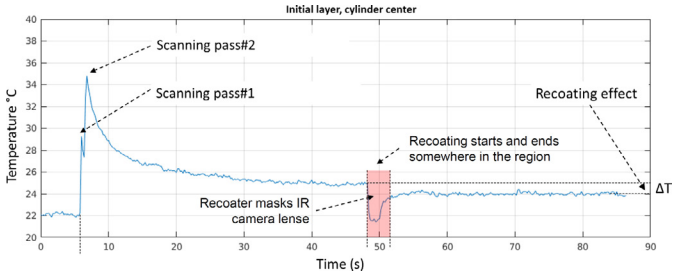


Fig. 3: Measured temperature evolution of a point in the spatial domain from photo-polymerisation until some time after re-coating. The very first peak of temperature is due to the very fast dynamics of photo-polymerisation while the cooling phase is much slower.

3. Initial physical model

As SLA is driven by a thermal process, the first model to be considered is the heat equation,

$$\partial_t(\rho \cdot c_p T) = \nabla \cdot (k \nabla T) \tag{1}$$

where ρ, c_p and k are the material density, specific heat capacity and conductivity. The material density and specific heat capacity are assumed to be stationary and the material conductivity is isotropic. Even though these assumptions cannot be verified a priori, the results from this work show that they are acceptable for our interest.

The IR camera does not allow the measurement of the temperatures inside the material. Therefore, the choice is made, at least initially, to derive the model using two spatial coordinates: the classic linear heat equation is obtained as a first physical model,

$$\frac{\partial T}{\partial t} = \alpha \cdot \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \tag{2}$$

where the diffusion coefficient is $\alpha = k/(\rho \cdot c_p)$. According to [1], meaningful physical characteristics of our material are $\rho \approx 1250$ kg/m3 and $c_p \approx 1190$ J/kg.K. The linear heat equation has analytical solutions, but since it was of our interest to modify the assumptions and potentially use a nonlinear model in the future, a numerical method that gives us some flexibility was chosen to solve it, namely finite differences. The initial conditions were chosen from experimental data and the boundary conditions according to Figure 4.

Some simulation results from the initial model are presented in Figures 7 and 8. As can be observed, the residual error grows quickly in the first 3 seconds, exhibiting the unmodelled fast dynamics of photo-polymerisation. A purely data-driven model is derived using a physics informed neural network, and simulation results show an important improvement over the initial physical model, see the black lines in Figure 8. The next section explains how these powerful models can be derived, as well as some of their limitations.

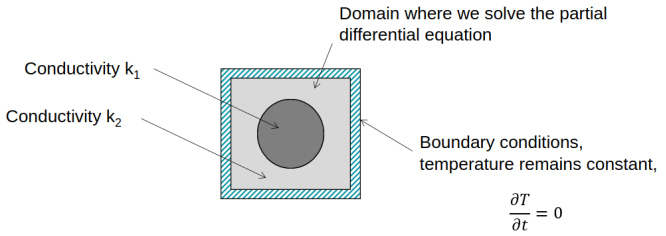


Fig. 4: Modeling assumptions: for the initial physical model, the material is considered homogeneous ($k_1 = k_2$) while for the improved model $k_1 \neq k_2$. The initial and boundary conditions are identical.

4. Improving the initial model using Physics Informed Neural Networks

Physics Informed Neural Networks (PINNs) have been recently introduced to discover partial differential equations from data, see [7] and [8]. Compared with more traditional techniques such as symbolic regression and sparse regression [9], PINNs have shown promising results even in the presence of noisy data [10]. In essence, PINNs are neural networks (NNs) with embedded physical knowledge: first by the choice of their inputs and second by the regularization terms used in their loss functions. For instance, in our case, the inputs of the deep neural network DNN_2 are chosen to be second order *spatial* derivatives, see Figure 5, but other spatial derivatives and/or polynomial terms could have been chosen. *Timewise*, if a wave equation would have been involved in the problem, $\hat{g}$ would have to be minimized to match a second order partial derivative in *time* rather than a first order. In our case, the first partial derivative in time is used in the loss function used to train DNN_3,

$$\mathcal{L} = \frac{1}{n} \left\| T_e - \hat{T} \right\|^2 + \frac{1}{n} \left\| \frac{\partial \hat{T}}{\partial t} - \hat{g} \right\|^2 + \frac{1}{n_0} \left\| T_{e0} - \hat{T}_0 \right\|^2 + \frac{1}{n_b} \left\| T_{eb} - \hat{T}_b \right\|^2 \tag{3}$$

where T_e are the experimental data points, T_{e0} are the experimental data points at $t = 0$ (initial conditions), T_{eb} are the experimental data points at the borders of the computational domain

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(boundary conditions), $\hat{T}$ is the output of the DNN_3, and $\hat{g}$ is the output of the DNN_2 and corresponds to the right-hand side of the partial differential equation (PDE) to be discovered.

The activation functions used in the NNs are sinus functions such that the derivative of a PINN is also a differentiable PINN and guaranteed to be smooth. In Figure 5, the three DNNs are trained sequentially using the L-BFGS-B optimizer and around 200,000 data points randomly sampled from the available data.

A major drawback of models derived using PINNs is that they are black-boxes in which one can have little trust. However, the last network trained, DNN_3, is a continuous function that fits the experimental temperature well while at the same time its derivatives are also fitted, see the loss function in Equation (3). This property is exploited to infer how the initial physical model could be improved.

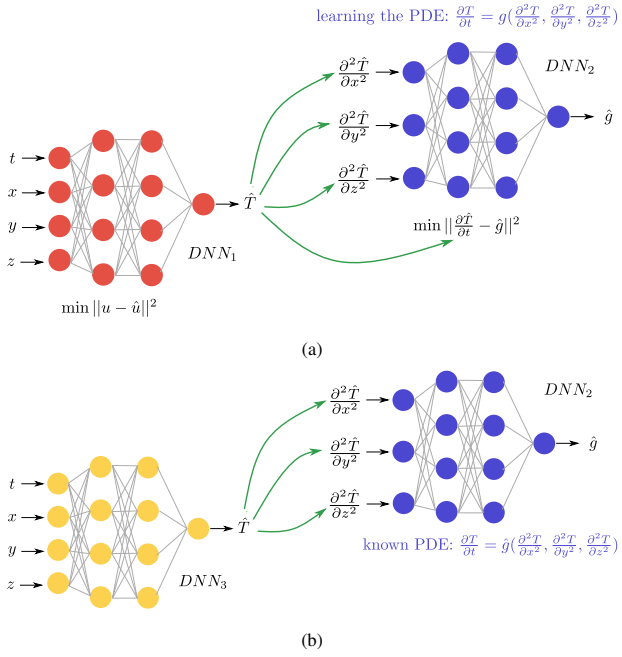


Fig. 5: Physics Informed Neural Networks: in (a) DNN_1 approximates the data while DNN_2 learns the left hand side of a PDE $g = \partial_t T$ as a function of chosen spatial derivatives, here represented $\partial_{xx}T, \partial_{yy}T$ and $\partial_{zz}T$. In (b) DNN_3 is fitted using the (trained) DNN_2, which represents the learned PDE and acts as a regularizer for DNN_3, see Equation (3). *Note: in this work only the two coordinates x and y where used.*

4.1. Model improvement

The accuracy of the partial derivatives obtained by first fitting neural networks and then performing automatic differentiation was shown to be quite superior [7] compared to more classical numerical differentiation methods. An illustration is depicted in Figure 6, where a forward simulation of the physics informed neural network DNN_3 trained directly on experimental data is presented. The trends obtained could not be reproduced using simple numerical differentiation such as central differences on the unfiltered or filtered noisy experimental data. These exhibit a pattern that we approximate as two different linear models with two diffusion coefficients whether inside or outside the area subjected to photo-polymerisation (solidification). This idea is used as a guide to improving the initial physical model which becomes

$$\partial_t T = \alpha_i \cdot \nabla^2 T \tag{4}$$

for $i = 1, 2$ with α_1 the diffusion coefficient inside the geometry under consideration and α_2 outside. From a physical point of view, we know from literature [1] that the diffusion coefficients should be the same ($\alpha_1 = \alpha_2$). From this, we conclude that the initial 2D approximation misses crucial thermal flows in z-direction. However, this quantity is hardly measurable, motivating the use of a model with nonphysical parameters. In that respect, the improved model proposed can be defined as hybrid as its structure is motivated by physics while the choice of its parameters is motivated by a data-driven approach.

An estimation of the diffusion coefficients is obtained by minimizing the residual temperatures,

$$\hat{\alpha} = \text{argmin}_\alpha \left(\frac{1}{n} \|T_e - T_s(\alpha)\|^2 \right) \tag{5}$$

where $T_s \in \mathbb{R}^n$ is obtained by solving Equation (4) using finite differences for fixed values of α_1 and α_2. $T_s(\alpha)$ is a nonlinear function, so the minimization problem is a nonlinear least-squares problem solved numerically using a gradient descent based solver. From a physical point of view, the values obtained for α_1 are 3 times higher than one would expect from a typical material [1]; α_2 is found, as expected, close to 0. Such a model and its parameters can be identified in less than 30 seconds on a low-end CPU (Processor Intel Xeon CPU E3-1545M v5 @ 2.90GHz, 2901 MHz, 4 Cores, 8 Logical Processors). Even though the model parameters lose their physical meaning, the accuracy of the model shows promising results as presented in the next section.

5. Numerical simulation results and discussions

The three models presented in the previous sections are compared in Figures 7 and 8. As a measure of the accuracy of the models, the maximum residual errors w.r.t. experimental measurements are computed for a couple of geometries. Overall, the maximum residual errors are below 3.5 degrees Celsius which is already interesting from an industrial point of view. The PINN black-box model is always a better approximation of the temperature field, when $t < 30$s. The proposed model, which is interpretable, performs between the PINN model and the initial physical model. Interestingly, both physical models have a hard time capturing some fast dynamics after the initial conditions. This might be due to the fact the initial conditions were chosen from the experimental data, and these might still contain some energy input from photo-polymerisation and not

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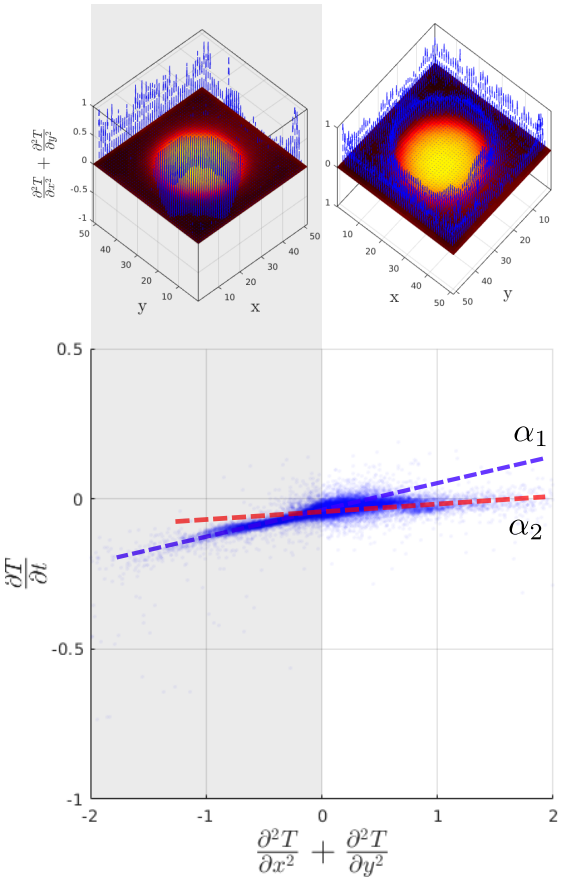
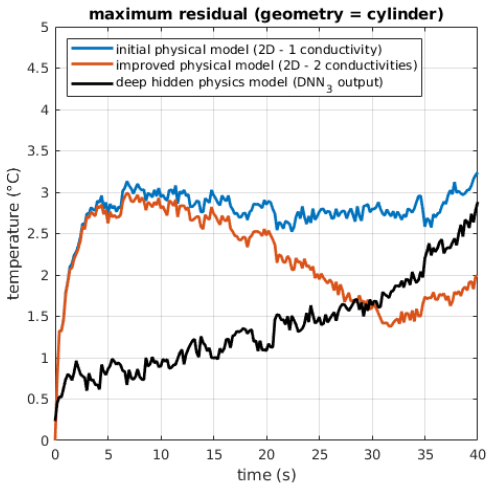
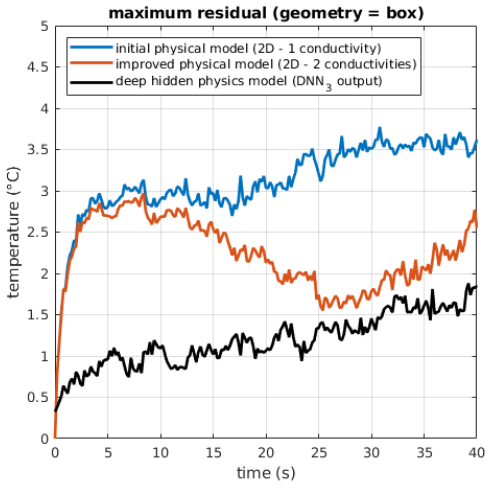


Fig. 6: A forward simulation of the PINN that is regularized using the learned PDE is performed and its partial derivatives are evaluated using automatic differentiation. Two different slopes (*diffusion* coefficients α_i) appear, suggesting an improvement to the initial physical model. The different slopes correspond to meaningful regions of the geometry (see top 3D plots at fixed t): α_1 inside the photo-polymerisation region has to be much higher than α_2 outside.



(a)



(b)

Fig. 7: Residual errors from the different models for a couple of geometries: (a) a cylinder and (b) a box. In blue, the initial physical model is always the worst. In black, the PINN black-box model is always a better approximation when $t < 30$s. The improved model has the advantage of being a white box model with an improved accuracy with respect to the initial model. Its parameters can also be evaluated at low cost compared to the PINN model.

only pure diffusion. Figure 8 shows the residuals are not homogeneous in space. The residual error of the improved physical model shows an interesting pattern: inside the cylinder the darker areas are the first ones to be scanned by the laser, meaning they had more time to diffuse — which probably explains some of the spatial distribution of the residual.

Computational cost. As shown in Figures 7 and 8, the PINNs-based model has the advantage of being the most accurate. However, training the deep neural network that can predict the temperature evolution, namely DNN_3, takes around 30min on a GPU (Nvidia GeForce GTX 1080 Ti). On the other hand, the forward simulation of the improved physics-based model, expressed in Equation (4), takes around 10s on the low-end CPU. The loss of accuracy of the improved physics-based model w.r.t. PINNs-based model is counter-balanced by the fact quick predictions of the temperature distribution can be made.

Implementation. The PINN-based models are implemented in TensorFlow with a Python interface, while the physics-based models are implemented in MATLAB.

6. Conclusions and future work

Experimental tests have been conducted to investigate the temperature evolution over simple geometries printed by stereolithography (SLA). Starting from the heat equation, an improved thermal model for SLA is derived in this paper. It consists of a hybrid model as its structure is motivated by the underlying physics while the choice of its parameters is motivated by a data-driven approach using Physics Informed Neural Networks (PINNs). Once the parameters of the model, namely *diffusion* coefficients, are identified, the improved thermal model can be run fast since the computational time is low. This implies that (*i*) improved model gives more insight in the process, hence better (process control) design, and (*i*) improved model is more accurate, so there is still room for reducing waiting times, hence

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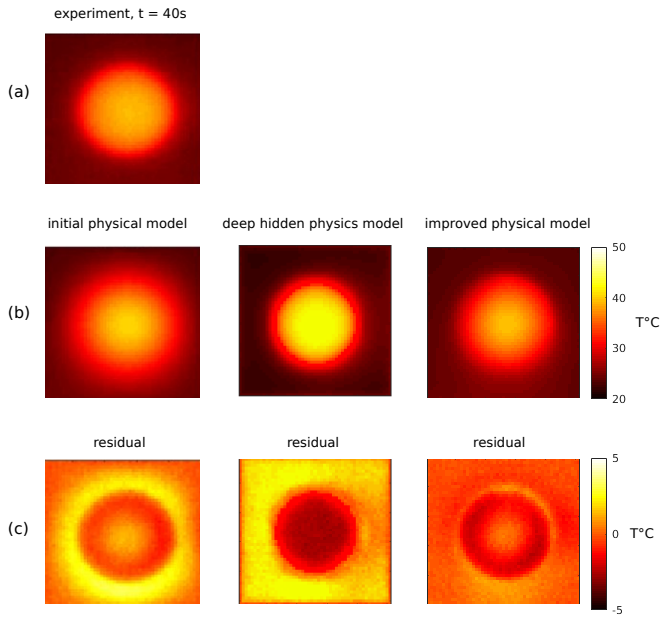


Fig. 8: Snapshots from (a) experimental data, (b) different model simulation results and in (c) the residuals (difference between (a) and (b) at t = 40s). The proposed model residual is the lowest one (see figure 7 (a)): here, one can see it is also the one with the most spatially homogeneous residual.

increasing production rate. In short, the improved model can be potentially used in industrial settings to improve the efficiency and productivity of SLA processes.

It would be interesting to add the third spatial dimension (i.e. vertical direction) to the physical model. However, a fundamental observability problem needs first to be solved, since the temperatures in this dimension are not measurable. Future work will investigate the use of a systematic way of deriving an interpretable model by employing sparse regression or evolutionary-based symbolic regression techniques to extract the equations from the black-box model captured by the deep neural network DNN_2. An additional PINN could be introduced into the previous workflow to model the material conductivity as a function of space to verify if the assumption about the material isotropy holds or not.

Since the technique is very generic, we will also apply it to other additive manufacturing techniques, e.g., laser-powder bed fusion (LPBF), posing major thermal problems.

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